

CERTIFIED RADIUS-SEVEN EXCHANGE RIGIDITY FOR A BINARY SUBSPACE CODE

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ABSTRACT. Let $\mathcal{C}_0$ be the 333-plane binary subspace code in $\mathbb{F}_2^7$ specified in Appendix C of Heinlein, Kiermaier, Kurz, and Wassermann. We certify that every code obtained by deleting at most seven planes from $\mathcal{C}_0$ and then adding arbitrary subspaces of arbitrary dimensions has at most 334 words, provided the minimum subspace distance is four. The proof reduces all exchanges to 2,561,563 unions of blocker sets. Each residual compatibility graph has at most 15 vertices and is solved by complete subset enumeration in an independent fail-closed verifier. The bound is attained by adjoining the full ambient space. This is a finite local rigidity result; it does not determine the global value of $A_2(7, 4)$, whose searched published frontier remains $334 \leq A_2(7, 4) \leq 388$.

1. INTRODUCTION

For subspaces U, W of a finite vector space, the subspace distance is

$$d_S(U, W) = \dim U + \dim W - 2 \dim(U \cap W).$$

Mixed-dimension subspace codes and their bounds are developed in [3]. In the binary seven-dimensional ambient space, Heinlein, Kiermaier, Kurz, and Wassermann constructed 333 three-dimensional codewords of minimum distance four [2]. Adjoining the full space gives a mixed-dimension code of size 334. Heinlein and Ihringer proved the published upper bound 388 by semidefinite programming [1]. The specialist table still displays the interval 334–388 [4].

A public exact replay had closed exchange radii through five around the 333-plane construction and listed radius six as a next experiment [5]. The present note advances that finite line of attack by closing radii six and seven. Its point is deliberately narrower than a claim about the global maximum: additions may have any dimension, but at most seven of the specified baseline planes may be deleted.

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2. THE BASELINE AND THE EXCHANGE PROBLEM

Put $V = \mathbb{F}_2^7$. Let $\mathcal{C}_0 = (P_0, \dots, P_{332})$ denote the code reconstructed from the two generators and 103 orbit representatives printed in Appendix C of [2]. We fix the canonical order obtained by sorting the binary membership fingerprints of the planes. Direct exact reconstruction gives the orbit census shown in Table 1.

TABLE 1. Reconstructed Appendix C orbit census.

Orbit length	1	2	4	Total planes
Number of orbits	9	26	68	$9 + 52 + 272 = 333$

All 333 objects have dimension three, are distinct, and have pairwise distance at least four. The independent ambient enumeration has layer sizes

$$1, 127, 2667, 11811, 11811, 2667, 127, 1,$$

whose sum is 29,212.

For an ambient subspace $X \notin \mathcal{C}_0$, define its *blocker set*

$$B(X) = \{i \in \{0, \dots, 332\} : d_S(X, P_i) < 4\}.$$

For $r \geq 0$, a radius- r exchange is any feasible code of the form

$$(1) \quad (\mathcal{C}_0 \setminus \{P_i : i \in R\}) \cup A, \quad |R| \leq r,$$

where A is an arbitrary set of subspaces outside $\mathcal{C}_0$. In particular, no constant-dimension restriction is imposed on A .

3. BLOCKER NORMALIZATION

Lemma 3.1 (Normalization). *If the exchange in (1) is feasible, then*

$$R' = \bigcup_{X \in A} B(X) \subseteq R,$$

and replacing R by R' preserves feasibility and cannot decrease the number of codewords.

Proof. Feasibility between an added subspace X and every retained baseline plane is equivalent to $B(X) \subseteq R$. Hence $R' \subseteq R$. A plane whose index belongs to $R \setminus R'$ is at distance at least four from every member of A , so restoring all such planes preserves every required pairwise distance and weakly increases cardinality. $\square$

It therefore suffices to consider removal sets that are unions of blocker sets. For such an R , write

$$E_R = \{X \notin \mathcal{C}_0 : B(X) \subseteq R\}.$$

The allowable additions are exactly the cliques in the distance-four compatibility graph on E_R . If that graph is G_R , the optimum for the fixed normalized removal set is

$$(2) \quad 333 - |R| + \omega(G_R).$$

4. EXACT ENUMERATION AND RESULT

The discovery computation represents a subspace by the 128-bit indicator of its vectors. It enumerates every canonical reduced-row-echelon basis, computes blockers by exact set intersection, closes the blocker masks under union up to the prescribed radius, and evaluates (2) by an exact coloring-based branch-and-bound.

At radius seven, 14,146 outside candidates have at most seven blockers, with the distribution in Table 2. Several candidates may have the same blocker set, leaving 14,140 distinct blocker sets. Their bounded union closure contains 2,561,563 removal sets. The eligible set E_R never has more than 15 vertices.

TABLE 2. Exact radius-seven enumeration.

Size	0	1	2	3	4	5	6	7
Candidates	1	12	16	416	1784	3087	4088	4742
Normalized removals	1	6	31	532	4738	32283	268060	2255912
Best code size	334	333	333	333	334	333	333	333

Theorem 4.1. *The maximum cardinality among all radius-seven exchanges around $\mathcal{C}_0$ is 334. In particular, no 335-word code can be obtained by deleting at most seven planes of $\mathcal{C}_0$ and adding arbitrary mutually compatible subspaces of V .*

Proof. Lemma 3.1 shows that it is enough to inspect every union of blocker sets of cardinality at most seven. The exact enumeration contains all 2,561,563 such unions. For each one, complete evaluation of the graph G_R gives the last row of Table 2, whose maximum is 334. Conversely, for $R = \emptyset$, adjoining the full space V to $\mathcal{C}_0$ gives 334 words: each baseline word has dimension three and hence is at distance $7 - 3 = 4$ from V . Thus the upper value in the finite search is attained. $\square$

The entry 334 at removal size four is a second equality witness: four planes are replaced by five compatible subspaces. It is recorded explicitly in the serialized result, although no equality classification is claimed.

5. INDEPENDENT VERIFICATION

The acceptance computation imports none of the discovery implementation or its outputs as trusted mathematical values. It rebuilds the Appendix C

orbits, rejects singular generators and overlapping orbits, and enumerates the 29,212 ambient subspaces independently. It recomputes distance by

$$d_S(U, W) = 2\text{rank}(U + W) - \dim U - \dim W,$$

rather than by membership intersection. Its blocker-union closure uses a split-union index different from discovery. Finally, for each graph of at most 15 vertices it tests all vertex subsets, at most 2^{15} , and accepts a subset precisely when it is a clique. Thus neither floating-point arithmetic nor an external optimizer occurs in the decisive verification.

The verifier is fail closed: exact JSON key sets, canonical full-rank RREF bases, data digests, all recomputed census fields, trace digest, and explicit equality witnesses must agree. Destructive tests confirm rejection of a duplicate codeword, an unknown result field, a singular group generator, malformed JSON, and a forged optimum of 335.

6. LIMITATIONS AND REPRODUCIBILITY

Theorem 4.1 is local to the specified $\mathcal{C}_0$. It does not exclude a distant 335-word clique, does not improve the published global upper bound 388, and does not exactify the semidefinite computation underlying that upper bound. A separate exact plane-packing MILP was attempted as a global constant-dimension diagnostic; its timed run did not close a useful bound and is not part of the proof.

The machine-readable input, discovery result, independent verifier, mutation tests, and SHA-256 release manifest accompany this paper. From the project root, the decisive commands are

```
python3 verification/verify_radius6.py --radius 7 \
  --result experiments/radius7_search_diagnostic.json
python3 tests/test_fail_closed.py --full
```

The computations use exact integer arithmetic over $\mathbb{F}_2$ and no randomness. No proof assistant was used. An AI coding agent assisted with literature retrieval, implementation, and manuscript drafting; every retained claim is bounded by the serialized certificate and the independent checks described above.

REFERENCES

1. Daniel Heinlein and Ferdinand Ihringer, *New and updated semidefinite programming bounds for subspace codes*, Advances in Mathematics of Communications **14** (2020), no. 4, 613–630.
2. Daniel Heinlein, Michael Kiermaier, Sascha Kurz, and Alfred Wassermann, *A subspace code of size 333 in the setting of a binary q -analog of the Fano plane*, Advances in Mathematics of Communications **13** (2019), no. 3, 457–475.
3. Thomas Honold, Michael Kiermaier, and Sascha Kurz, *Constructions and bounds for mixed-dimension subspace codes*, Advances in Mathematics of Communications **10** (2016), no. 3, 649–682.
4. Subspace Codes database, *Bounds for binary mixed-dimension subspace codes*, 2026, Table consulted August 30, 2026.

5. TheoremDB contributors, *Exact mixed-dimension subspace-code number in the binary 7-space at distance 4*, 2026, Statement P2796 and records R497, R500, and R504; snapshot reviewed July 28, 2026.

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